Chapter 2

FROM ROBOTS TO COBOTS

Machines and Robots are both integral to the world of technology and automation, yet they serve distinct purposes and exhibit contrasting characteristics

Difference between a machine and a robot

Some critical distinction between Robot and Machines :

Function and Purpose

Machines are typically designed with their specific task or function in mind. Machines are optimised to perform their designated task efficiently and effectively. In contrast, Robots have border range of functions and purposes. They can be programmed to perform various tasks such as assembly, welding, packaging and inspection or even complex activities like autonomous exploration or human like interactions

Automation and Autonomy

Machines are often operated by human users who control their action and monitor their performance. They rely on human inputs and instructions to carry out task. In contrast, Robots can operate autonomously or with minimal human intervention by continuously sensing their environment. They can execute predefined programs or algorithms and make decisions based on sensor inputs or environmental feedback

Flexibility and Adaptability

Machines are typically designed for specific applications or task and may have limited flexibility. They are often designed and optimised for a specific workflow or production process. In contrast, Robots are inherently more flexible and adaptable. They can be programmed or reprogrammed to perform different tasks and some advanced robots can even learn from experience or be trained to perform new task

Interaction and Perception

Machines usually interact with their environment in a predictable and predefined manner. They typically respond to specific inputs or triggers to perform their task. In contrast, Robots are equipped with sensors that allow them to perceive and interact with their environment more dynamically. They can use sensors such as cameras, depth sensors, Force sensors or tactile sensors to detect and respond to objects, people or events in their surroundings.

Complexity and Intelligence

Machines are often designed with a focus on mechanical or electrical functionality. They operate based on predefined rules, algorithms or mechanical interactions. Their complexity lies in optimising the physical design and mechanical components for their intended purpose. In contrast, Robots can incorporate advanced technologies and

capabilities beyond mechanical functionality. They can employ artificial intelligence ,techniques like machine learning or computer vision to enhance their decision making perception and adaptability

Using Examples

There are so many bulky things that have to move from one place to another in factories and industries. If a machine like a forklift may be used, with minimal human involvement, one person has to control and operate the vehicles (forklift) to lift the object. He has to pay attention to obstacles that come in between the path. He should carefully deliver it from one point to another.

On the other hand, when using an autonomous indoor vehicle with a robotic arm, then only one work left behind is to give the command to that Robot. The Robot will pick up the item and deliver it to the given place. The Robot automatically picks the item, plans the navigation route, avoids the course's obstacles and deliver it to the point.

Identification of machines and robots through illustrations

Autonomous operations

Many robots and machines are designed for autonomous operation, meaning they can perform task without constant human control. However not all robots and machines are autonomous. S some robots and machines may require human input or control to carry out their task

Physical appearance

Some robots are designed to be humanoid in appearance but many robots have non humanoid form. Robots can come in various shapes and sizes depending on their intended task and environments.

Purpose

Both robots and machines are designed for specific purpose and task. The distinction between them is not solely based on the specificity of their task. Robots can be designed for various applications beyond manufacturing and assembly such as exploration, research, medical assistance and more.

COBOTS

Cobots are known as collaborative robots. They are a type of robotic system designed to work alongside humans in a shared workspace. Cobots are also equipped with safety features and advanced sensors that allow them to interact safely with people

Difference between Robots and Cobots

Interaction - Robots work independently while Cobot interact with humans

Safety - Robots requires separation from Humanswhereas Cobots have built in safety features

Programming - Robots often require complex programmingwhereas Cobots are desined for easy programming and intuitive operation

Flexibility - Cobots are more adaptable to changing task and environment

Importance of Cobots

Safety - Cobots are designed to be safe to work with even in closed proximity to humans. This is achieved trough its slow speed, soft-stop technology and physical barriers

Flexibility and Adaptability - Cobots are generally more flexible and adaptable compared to traditional Robots. They are often light weight, portable and easy to reconfigure, allowing for quick depyment and efficient use in different work environments

Ease of Programming - Cobots often have user friendly programming interfaces that simplify programming and re programming process

Benefits of using Cobots in manufacturing

Improved Safety - Cobots are deigned to work alongside humans reducing risk of accidens and injuries. They are equipped with sensors and saftey features to prevent collisions

Increased efficiency - Cobots automate repetitive and mundane task, freeing up human workers for more complex and high value task

Enhanced Quality - Cobots provide consistent and precise performance, reducing the likelihood of human error and improving overall product quality

E**nhanced Ergonomics**

Cobots are designed with human ergonomics in mind. They can handle physically demanding or repetitive task, reducing the strain and fatigue on human workers

Cost effectiveness

Cobots are typically less expensive than traditional industrial robots. This makes them a good option for small business and startups that want to automate their operation without breaking the bank.

Collaboration

Cobots are specifically designed for collaboration with humans. Unlike traditional Robots that are typically operated by skilled technicians or require programming expertise, Cobots are often easy to program and can be taught or guided by non experts.

Progress from Robots to Cobots

A Cobot is a Robot designed to work safely alongside humans. It is a device and method for direct physical interaction between a person and a computer controlled manipulator

Cobots possess the innate ability to work in tandem with humans, whereas traditional industrial robots work in place of humans

Cobots can fulfill various roles. They can take over activities that employees experience as ‘boring’ and ‘monotonous’

The advantages of using Cobots are, that robots no longer have to be in cages or safety zones.

Cobots are equipped with force feedback sensors that prevent them from injuring colleagues. They are also easier to program and can be used more flexibily than industrial robots